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RESEARCH TOPIC

Advances in CRISPR Cas gene editing and implications for rare genetic diseases

EXECUTIVE SUMMARY

Topic Framing

The application of CRISPR-Cas gene editing technology has shown significant promise in addressing rare genetic diseases, which often result from specific genetic mutations. These diseases pose substantial challenges due to their complexity and the lack of effective treatments. The retrieved articles provide insights into the current advances in CRISPR-Cas technology and its implications for treating rare genetic disorders, with a focus on collagen disorders and the broader landscape of rare diseases in regions like India.

Most Relevant Papers and Contributions

1. **"Gene editing for collagen disorders: current advances and future perspectives"** highlights the potential of CRISPR-Cas systems in treating collagen disorders, which are characterized by a wide range of clinical features due to pathogenic variants in collagen genes. The paper underscores the promise of CRISPR-Cas in providing therapeutic solutions for these complex conditions, which affect multiple systems such as musculoskeletal, cardiovascular, and respiratory systems.
2. **"Rare genetic disorders in India: Current status, challenges, and CRISPR-based therapy"** discusses the prevalence of rare genetic diseases in India and the potential of CRISPR-Cas9 technology to offer precise, mutation-specific treatments. This review emphasizes the challenges faced in implementing these therapies, including national policy constraints and healthcare infrastructure limitations.
3. **"Detection of pre-existing immunity to bacterial Cas9 proteins in people with cystic fibrosis"** addresses the challenges of immune responses to CRISPR-Cas9, particularly in the context of cystic fibrosis. This paper highlights the importance of understanding immune reactions to gene editing tools, which is crucial for the successful application of CRISPR-based therapies in complex in vivo environments.

Recurring Trends and Methodological Patterns

- **Precision and Specificity:** CRISPR-Cas technology is celebrated for its precision in targeting specific genetic mutations, making it particularly suitable for monogenic disorders. This precision is a recurring theme across the literature, with multiple papers emphasizing its potential to correct disease-causing mutations effectively.
- **Challenges in Immune Response:** The issue of immune responses to CRISPR components, such as Cas9 proteins, is a significant concern. The detection of pre-existing immunity in cystic fibrosis patients underscores the need for strategies to mitigate immune reactions, which is a critical consideration for the clinical application of CRISPR therapies.
- **Regional and Policy Considerations:** The implementation of CRISPR-based therapies is influenced by regional healthcare policies and infrastructure, as highlighted in the context of India. This suggests that while the technology holds promise, its deployment is contingent on overcoming systemic healthcare challenges.

Limitations and Evidence Gaps

- **Immune System Interactions:** There is a notable gap in understanding the full extent of immune responses to CRISPR components, which could impact the safety and efficacy of gene editing therapies. Further research is needed to develop strategies to circumvent these immune challenges.
- **Clinical Translation:** While the potential of CRISPR-Cas for treating rare genetic diseases is well-documented, the transition from laboratory research to clinical application remains a significant hurdle. This includes addressing ethical, regulatory, and technical challenges associated with gene editing in humans.
- **Diverse Genetic Landscapes:** The variability in genetic mutations across different populations and the complexity of diseases like collagen disorders require tailored approaches. The current literature suggests a need for more comprehensive studies that consider genetic diversity and its implications for CRISPR-based treatments.

In conclusion, while CRISPR-Cas technology represents a transformative advancement in the treatment of rare genetic diseases, several challenges must be addressed to realize its full potential. These include managing immune responses, navigating regional healthcare systems, and ensuring the safe and ethical application of gene editing technologies.

1

Gene editing for collagen disorders: current advances and future perspectives.

PubMed

2025-08-11

DOI: 10.1038/s41434-025-00560-7

Abstract. Collagen disorders encompass a wide range of genetic conditions caused by pathogenic variants in collagen genes for which there is an unmet need for treatments. They present various clinical features, ranging from localised tissue abnormalities to severe systemic complications. Symptoms differ significantly and depend on the pathogenic variant, which can affect various systems, including the musculoskeletal, cardiovascular, and respiratory systems, highlighting the complex implications of collagen gene pathogenic variants and the wide range of expression patterns among different collagen types. Gene-editing technologies, particularly Clustered Regularly Interspaced Palindromic Repeats (CRISPR)-Cas systems, have emerged as promising therapeutic options for these disorders, representing a putative one-for-all treatment strategy. This review provides an overview of current gene-editing strategies aimed at collagen-related diseases, including osteogenesis imperfecta, Alport syndrome, and dystrophic epidermolysis bullosa. We explore the application of CRISPR-Cas9, which facilitates targeted DNA modifications, base editing (BE), and prime editing (PE), enabling precise single-nucleotide

Source URL. <https://pubmed.ncbi.nlm.nih.gov/40790091>

2

Rare genetic disorders in India: Current status, challenges, and CRISPR-based therapy.

PubMed

2024-02-01

Abstract. Rare genetic diseases are a group of life-threatening disorders affecting significant populations worldwide and posing substantial challenges to healthcare systems globally. India, with its vast population, is also no exception. The country harbors millions of individuals affected by these fatal disorders, which often result from mutations in a single gene. The emergence of CRISPR-Cas9 technology, however, has ushered in a new era of hope in genetic therapies. CRISPR-based treatments hold the potential to precisely edit and correct diseasecausing mutations, offering tailored solutions for rare genetic diseases in India. This review explores the landscape of rare genetic diseases in India along with national policies and major challenges, and examines the implications of CRISPR-based therapies for potential cure. It delves into the potential of this technology in providing personalized and effective treatments. However, alongside these promising prospects, some ethical considerations, regulatory challenges, and concerns about the accessibility of CRISPR therapies are also discussed since addressing these issues is crucial for harnessing the full power of CRISPR in tackling rare gene

Source URL. <https://pubmed.ncbi.nlm.nih.gov/38383973>

3

Advances in gene editing for legume improvement: technologies, progress, and prospects

OpenAlex

2026-04-15

DOI: <https://doi.org/10.3389/fgeed.2026.1789952>

Abstract. Legumes are among the most important protein-rich crops in global agri-food systems. To meet the rising protein demand of a growing population, significant efforts are underway to enhance legume yield, nutritional quality, and resilience to environmental stresses through the manipulation of key genetic traits. Advanced technologies-including genetic engineering, gene editing, genomic selection, next-generation sequencing, single-cell genomics, and multi-omics-are accelerating legume improvement due to their high precision and efficiency. This review focuses on major gene-editing technologies, namely, CRISPR/Cas9 (Clustered Regularly Interspaced Short Palindromic Repeats/CRISPR-associated protein 9), TALENs (Transcription Activator-Like Effector Nucleases), ZFNs (Zinc Finger Nucleases), base editing (BE), and prime editing (PE), and their applications in key legume crops such as soybean (*Glycine max*), cowpea (*Vigna unguiculata*), chickpea (*Cicer arietinum*), groundnut (*Arachis hypogaea*), pea (*Pisum sativum*), barrel clover (*Medicago truncatula*), alfalfa (*Medicago sativa*), and *Lotus japonicus* . Among these platforms, CRISPR/Cas9 is the most widely adopted in legumes, large

Source URL. <https://openalex.org/W7154509363>

4

Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-Cas Unleashed: Transforming Gene Editing With Breakthroughs, Applications, and Ethical Dilemmas.

EuropePMC

2025-11-01

DOI: [10.7759/cureus.95908](https://doi.org/10.7759/cureus.95908)

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41211267>

5

CRISPR/Cas9 Genome Editing in Rice (*Oryza sativa* L.) for Multi-Stress Resilience and Prospects for Agricultural Sustainability

OpenAlex

2026-04-16

DOI: <https://doi.org/10.1007/s10343-026-01328-8>

Abstract. Rice (*Oryza sativa* L.) is a staple food for nearly half of the world's population, making it vital for food security and economic stability. Currently, severe climate change poses serious threats to rice production, as both abiotic and biotic stresses cause significant yield losses. To meet these challenges, developing innovative strategies to create resilient rice varieties is crucial. CRISPR/Cas9 genome editing has become a transformative technology for precise, rapid, and transgene-free improvement of complex stress tolerance traits in rice. This review offers a comprehensive overview of recent advances in CRISPR/Cas9-mediated genome editing aimed at enhancing multi-stress resilience in rice against stresses. It holds significant potential for sustainability by supporting microbiome engineering and biofortification strategies that can sustain long-term global food and nutritional security. It promotes the development of rice cultivars rich in essential nutrients, resistant to diseases, and adaptable to changing climate conditions. Additionally, this review includes evidence from the analysis of homologous genes between *Arabidopsis thaliana* and rice to identify functionally conse

Source URL. <https://openalex.org/W7154590301>

6

Detection of pre-existing immunity to bacterial Cas9 proteins in people with cystic fibrosis.

PubMed

2025-09-01

DOI: 10.1093/immhor/vlaf041

Abstract. Cystic fibrosis (CF) is caused by homozygous mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene, resulting in multi-organ dysfunction and decreased lifespan and quality of life. A durable cure for CF will likely require a gene therapy approach to correct CFTR. Rapid advancements in genome editing technologies, including CRISPR/Cas9, have already resulted in Food and Drug Administration (FDA) approval for cell-based gene editing therapies, providing new therapeutic avenues for many rare diseases. However, immune responses to gene therapy delivery vectors and editing tools remain a challenge, especially for strategies targeting complex in vivo tissues such as the lung. Previous findings in non-CF healthy individuals reported pre-existing antibody and T cell responses to recombinant Cas9 proteins, suggesting potential additional obstacles for incorporation of clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9 technologies in gene therapies. To determine whether pre-existing immunity to Cas9 from *S. aureus* or *S. pyogenes* was present or augmented in people with CF, anti-Cas9 IgG levels and Cas9-specific T cell responses were determined

Source URL. <https://pubmed.ncbi.nlm.nih.gov/40972653>

7

Modelling human lower urinary tract malformations in zebrafish.

PubMed

2023-03-29

DOI: 10.1186/s40348-023-00156-4

Abstract. Advances in molecular biology are improving our understanding of the genetic causes underlying human congenital lower urinary tract (i.e., bladder and urethral) malformations. This has recently led to the identification of the first disease-causing variants in the gene *BNC2* for isolated lower urinary tract anatomical obstruction (LUTO), and of *WNT3* and *SLC20A1* as genes implicated in the pathogenesis of the group of conditions called bladder-exstrophy-epispadias complex (BEEC). Implicating candidate genes from human genetic data requires evidence of their influence on lower urinary tract development and evidence of the found genetic variants' pathogenicity. The zebrafish (*Danio rerio*) has many advantages for use as a vertebrate model organism for the lower urinary tract. Rapid reproduction with numerous offspring, comparable anatomical kidney and lower urinary tract homology, and easy genetic manipulability by Morpholino®-based knockdown or CRISPR/Cas editing are among its advantages. In addition, established marker staining for well-known molecules involved in urinary tract development using whole-mount in situ hybridization (WISH) and the usage of transgenic lines expressing fluor

Source URL. <https://pubmed.ncbi.nlm.nih.gov/36977792>

8

Epigenome editing strategies: technological progress and in vivo applications

OpenAlex

2026-04-14

DOI: <https://doi.org/10.1186/s44330-026-00069-4>

Abstract. Abstract Genome editing refers to a group of technologies that allow targeted modifications at specific sites within the genome. Advances in these technologies have not only made it possible to precisely alter the genome but also to manipulate the epigenome of specific regions, allowing for the controlled regulation of gene expression. It is increasingly evident that the epigenome plays a crucial role in the regulation of gene expression during development and differentiation, and that epigenome abnormalities are linked to various diseases including cancer and lifestyle-related disorders. As a result, the ability to manipulate the epigenome of a specific region offers valuable insights into its function in development and differentiation and also presents potential for therapeutic interventions across a broad spectrum of disease. Given its growing significance, epigenome editing is poised to become a critical tool in both basic and medical research. In this review, we provide a comprehensive overview of the latest advancements in epigenome editing technologies.

Source URL. <https://openalex.org/W7154264165>

9

Bridging science and hope: the evolving story of gene therapy for neuromuscular diseases

OpenAlex

2026-04-14

DOI: <https://doi.org/10.3389/fcell.2026.1765367>

Abstract. The field of gene therapy for neuromuscular dystrophies has evolved over the past two decades. Despite some outstanding positive outcomes, some unfortunate adverse effects also led to big setbacks. One important key point is to study relevant preclinical models and to embrace diverse strategies to mitigate or avoid such negative outcomes. Although at first, for some diseases, the promise of a one-treatment-for-all approach was envisioned, it has recently become clear that a personalized approach will likely be preferable given the high variability in response between individuals.

Source URL. <https://openalex.org/W7154288389>

10

Advances in SRNS Gene Research: From Precision Classification to Precision Diagnosis and Treatment.

EuropePMC

2026-03-19

DOI: [10.3390/biomedicines14030711](https://doi.org/10.3390/biomedicines14030711)

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41898355>

11

Gene, genetics and genetic medicines in gastroenterology: Current status and its future.

EuropePMC

2026-01-01

DOI: 10.3748/wjg.v32.i1.112496

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41551533>

12

Expanding the CRISPR/Cas toolkit: applications in proteomics and theranostics.

EuropePMC

2025-11-27

DOI: 10.3389/fbioe.2025.1713700

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41394966>

13

Genetic Disorders and its Prevention- An Ayurvedic Perspective

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.26463/rjas.3_1_6

Abstract. Genetics is one of the most interesting topics of research in the current medical science. This science of heredity has got an elaborate description in the authoritative books of Ayurveda. Modern genetics deals with the combination of maternal and paternal genes as the responsible factor for an offspring. Genetics explains how the gametes get mixed with each other and how a specific gene gets stimulated. But books in Ayurveda deals much more than this. Ayurveda has explained the role of different factors favoring this development such as - Matruja Pitruja Atmaja Satvaja Rasaja Satymaja which explains that the child born to a couple is not merely a blend of parental and maternal characters alone but a amalgamation of many more factors. Ayurveda is conceptually well equipped to explain that the constitutional make up of a progeny can be decided by scientific methods in line with the present day Genetic Engineering concepts. Proper following of Rtukalanyamas -Dos and donts at the time of Rtukala. avoiding Garbhopaghatakarabhavas -Factors hindering the growth of fetus. following of MasanumasikaParicharya -Month wise regimens for the pregnant women are few of them. Ayurveda advises pre

Source URL. https://doi.org/10.26463/rjas.3_1_6

14

The Evolving Landscape of Fetal Therapy: Surgical Interventions and Emerging Biologics

OpenAlex

2026-04-13

DOI: <https://doi.org/10.3390/biologics6020011>

Abstract. Fetal therapy has evolved into a rapidly advancing field with the potential to alter the natural history of many severe congenital and genetic disorders before irreversible injury occurs. Progress in prenatal imaging, molecular diagnostics, and fetal intervention techniques now enables the earlier identification of disease and, in select settings, targeted prenatal treatment. This review synthesizes the current landscape of fetal therapies, spanning established surgical interventions for structural anomalies and emerging biologic and molecular approaches, including enzyme replacement therapy, stem cell-based strategies, gene therapy, and gene editing. The intrauterine environment provides a distinct therapeutic context, with developmental plasticity, immune immaturity, enhanced tissue accessibility, and relatively permissive central nervous system exposure that together define a time-sensitive window for intervention. Preclinical studies and early clinical experience across both structural anomalies and genetic disorders, including lysosomal storage disorders, osteogenesis imperfecta, and spinal muscular atrophy, support the premise that prenatal treatment can preserve organ develop

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15

Primary high grade mucoepidermoid carcinoma of skin and cervical lymphatic affection a rare entity

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: [10.15406/mojs.2018.06.00136](https://doi.org/10.15406/mojs.2018.06.00136)

Abstract. No abstract available.

Source URL. <https://doi.org/10.15406/mojs.2018.06.00136>

16

Structural Identification of Large Finite Element Models Using Commodity Computing Clusters for Parallel Genetic Algorithms

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: [10.12783/shm2015/112](https://doi.org/10.12783/shm2015/112)

Abstract. No abstract available.

Source URL. <https://doi.org/10.12783/shm2015/112>

17

La face sombre de la motivation de service public : Analyse de la corruption pour une cause noble à partir du cas libanais

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.3917/gmp.pr1.0018

Abstract. Se fondant sur 32 entretiens avec des fonctionnaires libanais, pays touché par la corruption et une situation sociale difficile, le but de cet article est de comprendre comment la motivation de service public (MSP) conduit les fonctionnaires à développer une corruption pour une cause noble (CCN). Cette recherche est doublement exploratoire. Premièrement, s'intéresser aux effets négatifs de la MSP, c'est enrichir un champ de recherche essentiellement tourné vers la compréhension des effets positifs de ce concept. Deuxièmement, se focaliser sur la CCN c'est là encore mobiliser un concept qui jusqu'ici a été mobilisé pour le cas de policiers. Mobilisant les travaux sur les deux concepts et ceux sur la justification morale, cette recherche réalise une double contribution théorique. Premièrement, nous montrons que les dimensions de compassion et d'abnégation de la MSP conduisent à une CCN dont nous précisons les contours (soutien supplémentaire et protection contre une sanction). Cette dynamique est source de tensions internes pour le fonctionnaire qui choisit de réaliser une CCN, tant il est affecté par la vulnérabilité des usagers. Deuxièmement, nous dégagons trois facteurs contextuels

Source URL. <https://doi.org/10.3917/gmp.pr1.0018>

18

Transformation digitale des services publics locaux et valeur publique : une étude de cas sur la santé et la formation professionnelle en France

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.3917/gmp.pr1.0007

Abstract. Le lien entre digitalisation et création de valeur publique reste à explorer. Dans cette optique, cette recherche présente une analyse qualitative de 42 entretiens semi-directifs menés avec des acteurs publics et privés de la santé et de la formation professionnelle opérant dans 10 régions de France. Les résultats montrent que la digitalisation peut créer de la valeur publique lorsqu'elle donne lieu à (i) une reconfiguration des organisations publiques visant la prise en compte des besoins exprimés par les acteurs de terrain ; (ii) un mode de coordination favorisant l'apport des citoyens-usagers dans la création de nouveaux services ; et (iii) l'alignement des valeurs des parties prenantes impliquées dans la création de valeur découlant de ces services. Partant de ces résultats, trois leviers de création de valeur publique induite par la digitalisation sont proposés pour contribuer à la littérature et aider les acteurs publics locaux à s'engager avec plus de possibilités de succès dans les projets de transformation digitale.

Source URL. <https://doi.org/10.3917/gmp.pr1.0007>

19

Rare Adrenal Gland Emergencies: A Case Series of Giant Myelolipoma Presenting With Massive Hemorrhage and Abscess

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.5812/numonthly.22671

Abstract. No abstract available.

Source URL. <https://doi.org/10.5812/numonthly.22671>

20

Traitement de la symptomatologie dépressive concomitante à la maladie d'Alzheimer : étude de cas des effets d'une psychothérapie psychodynamique interpersonnelle

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.3917/bupsy.pr1.002

Abstract. Cette étude de cas avait pour objectif l'exploration des effets d'une psychothérapie psychodynamique interpersonnelle sur les symptômes dépressifs d'une participante (nom fictif : Clara) atteinte d'un trouble neurocognitif dû à la maladie d'Alzheimer (TNMA). Les outils de mesure utilisés ont inclus la Dementia Mood Assessment Scale (DMAS) et la Cornell Scale for Depression in Dementia (CSDD). L'utilisation d'une mesure continue de la détresse a permis d'apprécier la présence de changements significatifs tout au long de l'intervention. Le modèle de psychothérapie adopté a fait l'objet d'un approfondissement sur le plan de ses fondements théoriques reliés à la pratique. Le contenu des échanges a été analysé en fonction de différents outils et objectifs associés à ce modèle. Une amélioration significative a été rapportée par Clara à la CSDD et la mesure est poursuivie. Une baisse significative à la DMAS et à la CSDD a été notée par le proche aidant. En conclusion les résultats semblent appuyer la pertinence clinique et scientifique du modèle de psychothérapie proposé.

Source URL. <https://doi.org/10.3917/bupsy.pr1.002>

Approche psychopathologique de l'hyperactivité et pathologies limites chez l'enfant : le cas de deux jeunes filles

[CrossRef](#)

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.3917/bupsy.pr1.009

Abstract. L'hyperactivité et les difficultés attentionnelles sont un motif particulièrement fréquent de consultation psychologique de l'enfant. Dans la perspective neurocognitiviste du développement, dominante actuellement en pédopsychiatrie, ces symptômes sont généralement considérés comme des signes possibles d'un trouble neurodéveloppemental. Notre objectif est d'illustrer qu'ils peuvent aussi être décrits d'un point de vue psychodynamique comme l'expression d'une souffrance psychique liée à différentes psychopathologies. Sont présentées deux études de cas menées dans le cadre de bilans psychologiques avec deux jeunes filles de 6 et 7 ans souffrant de ce type de symptômes. Les données cliniques (entretiens, tests cognitifs, dessins, projectifs) concourent à montrer que leur hyperactivité est sous-tendue par un fonctionnement de type limite pour l'une et par une problématique dépressive pour l'autre. Cette recherche soutient qu'une démarche diagnostique complémentariste est fondamentale pour mener une conduite thérapeutique appropriée qui prenne en charge de manière complémentaire le symptôme et le conflit psychique.

Source URL. <https://doi.org/10.3917/bupsy.pr1.009>

Le rôle des effets de réseau locaux et des effets spatiaux dans le développement des plateformes numériques : le cas du site de covoiturage iDVROOM

[CrossRef](#)

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.3917/reco.pr2.0181

Abstract. L'objet de cet article est de comprendre le rôle des effets de réseau locaux et des effets spatiaux dans le développement des plateformes numériques, à travers l'exemple du site de covoiturage iDVROOM. Ce site est une plateforme biface au sens de Rochet et Tirole [2003], mais avec deux spécificités : 1) la plupart des utilisateurs sont sur les deux faces de la plateforme (à la fois conducteurs et passagers) et 2) les utilisateurs interagissent localement, le plus souvent au sein de communautés. Nous analysons l'impact du nombre d'utilisateurs inscrits et de communautés créées sur la croissance de la plateforme. À partir de données couvrant la période 2015-2018, nous montrons que l'activité de covoiturage dans une commune est stimulée à la fois par la création de communautés et par la base d'utilisateurs dans cette commune et dans les communes voisines. Toutefois, les effets de réseau sont modérés par le caractère local des interactions entre utilisateurs de la plateforme. Classification JEL : L14, L92, M13, R32.

Source URL. <https://doi.org/10.3917/reco.pr2.0181>

23

Expression and Polymorphism of the Progesterone-Associated Endometrial Protein Gene and Serum Parameters in Relation to Intracytoplasmic Sperm Injection Outcomes among Iraqi Women

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.21608/ejmm.2026.460581.2117

Abstract. No abstract available.

Source URL. <https://doi.org/10.21608/ejmm.2026.460581.2117>

24

Rare and very rare decays at the LHCb experiment

arXiv

2026-04-16

Abstract. Rare and very rare decays of third-generation particles, including b -hadrons and τ leptons, provide sensitive probes of physics beyond the Standard Model (SM). Unlike direct searches limited by collider energies, they probe new physics at much higher energy scales. Many of these decays have SM-predicted branching fractions below the sensitivity of current detectors. These proceedings report on recent LHCb searches, including several first searches and results setting the most stringent limits to date. In particular, searches for $b \rightarrow s \tau^+ \tau^-$, $b \rightarrow s \tau^\pm \mu^\mp$, $b \rightarrow s \mu^\pm \mu^\mp$, and $\tau^\pm \rightarrow \mu^\pm \mu^\pm \mu^\mp$ are presented, alongside searches for lepton-number-violating processes and loop-suppressed annihilation decays.

Source URL. <http://arxiv.org/abs/2604.15199v1>

25

Recent Advances in the Non-viral Delivery of Genes to Central Nervous System Disorders.

EuropePMC

2026-03-21

DOI: 10.1007/s10571-026-01703-z

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41865126>

26

From Bench to Bedside: Ethical and Clinical Best Practices for Genome Editing Applications.

EuropePMC

2026-02-02

DOI: 10.3390/ijms27031484

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41683905>

27

Molecular Crosstalk in Age-Related Macular Degeneration: Integrating Oxidative Stress, Inflammation, microRNAs, and Genetic Susceptibility Toward Precision Therapeutics

EuropePMC

2026-02-01

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/PMC/PMC12938228>

28

Drug-tolerant persister cells: the deadly survivors in hematological malignancies

OpenAlex

2026-04-14

DOI: <https://doi.org/10.3389/fphar.2026.1740281>

Abstract. Although the prognosis of patients with hematologic cancers has improved significantly owing to the advance of radiotherapy, chemotherapy, targeted therapies, immunotherapies, and hematopoietic stem cell transplantation therapy, disease relapse and refractory are still very prevalent and remain the major obstacles for long-term patient survival. Recent evidence suggests that drug-tolerant persister (DTP) cells are considered to be the cellular reservoir for tumor relapses and drug resistance formation. DTP cells refer to tumor cells that are acquired tolerance to anticancer drugs during exposure. Distinct from genetic mutations which cause stable drug-resistance, drug-tolerance in DTP cells is reversible which is primarily mediated by non-genetic alterations including changes in epigenetic modification, metabolic reprogramming, and/or transcriptional regulation. However, under prolonged or repeat drug-exposure, DTP cells acquire mutations in genes that are required for drug-treatment response and become stable drug-resistance. Targeting stable drug-resistant cell therapy is very challenge which needs novel drugs. Thus, targeted DTP cells therapy is particularly interesting which on

Source URL. <https://openalex.org/W7154209283>

29

TokenLight: Precise Lighting Control in Images using Attribute Tokens

arXiv

2026-04-16

Abstract. This paper presents a method for image relighting that enables precise and continuous control over multiple illumination attributes in a photograph. We formulate relighting as a conditional image generation task and introduce attribute tokens to encode distinct lighting factors such as intensity, color, ambient illumination, diffuse level, and 3D light positions. The model is trained on a large-scale synthetic dataset with ground-truth lighting annotations, supplemented by a small set of real captures to enhance realism and generalization. We validate our approach across a variety of relighting tasks, including controlling in-scene lighting fixtures and editing environment illumination using virtual light sources, on synthetic and real images. Our method achieves state-of-the-art quantitative and qualitative performance compared to prior work. Remarkably, without explicit inverse rendering supervision, the model exhibits an inherent understanding of how light interacts with scene geometry, occlusion, and materials, yielding convincing lighting effects even in traditionally challenging scenarios such as placing lights within objects or relighting transparent materials plausibly. Pro

Source URL. <http://arxiv.org/abs/2604.15310v1>

30

AnimationBench: Are Video Models Good at Character-Centric Animation?

arXiv

2026-04-16

Abstract. Video generation has advanced rapidly, with recent methods producing increasingly convincing animated results. However, existing benchmarks—largely designed for realistic videos—struggle to evaluate animation-style generation with its stylized appearance, exaggerated motion, and character-centric consistency. Moreover, they also rely on fixed prompt sets and rigid pipelines, offering limited flexibility for open-domain content and custom evaluation needs. To address this gap, we introduce AnimationBench, the first systematic benchmark for evaluating animation image-to-video generation. AnimationBench operationalizes the Twelve Basic Principles of Animation and IP Preservation into measurable evaluation dimensions, together with Broader Quality Dimensions including semantic consistency, motion rationality, and camera motion consistency. The benchmark supports both a standardized close-set evaluation for reproducible comparison and a flexible open-set evaluation for diagnostic analysis, and leverages visual-language models for scalable assessment. Extensive experiments show that AnimationBench aligns well with human judgment and exposes animation-specific quality differences overlooked

Source URL. <http://arxiv.org/abs/2604.15299v1>

31

Why Do Vision Language Models Struggle To Recognize Human Emotions?

arXiv

2026-04-16

Abstract. Understanding emotions is a fundamental ability for intelligent systems to be able to interact with humans. Vision-language models (VLMs) have made tremendous progress in the last few years for many visual tasks, potentially offering a promising solution for understanding emotions. However, it is surprising that even the most sophisticated contemporary VLMs struggle to recognize human emotions or to outperform even specialized vision-only classifiers. In this paper we ask the question "Why do VLMs struggle to recognize human emotions?", and observe that the inherently continuous and dynamic task of facial expression recognition (DFER) exposes two critical VLM vulnerabilities. First, emotion datasets are naturally long-tailed, and the web-scale data used to pre-train VLMs exacerbates this head-class bias, causing them to systematically collapse rare, under-represented emotions into common categories. We propose alternative sampling strategies that prevent favoring common concepts. Second, temporal information is critical for understanding emotions. However, VLMs are unable to represent temporal information over dense frame sequences, as they are limited by context size and the number

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New frontiers in quantum science and technology using van der Waals Josephson junctions

arXiv

2026-04-16

Abstract. Over the last decade, the development of Josephson devices based on van der Waals (vdW) materials has advanced rapidly, representing a paradigm shift driven by the advent of 2D materials. The diverse vdW materials library, combined with advanced fabrication techniques, enables the integration of materials with vastly disparate properties for scientific exploration. The vdW Josephson junctions (JJs) offer a unique route to explore novel functionalities and associated physics that remain inaccessible in conventional JJs, which have reached an industrial level in terms of fabrication. Beyond material diversity, vdW crystalline materials offer fundamental new control over device symmetries, enabling the realization of Hamiltonians unique to 2D systems. Furthermore, the long relaxation times of myriad excitations in 2D heterostructures open possibilities for creating exquisite quantum sensors, with the 2D material itself acting as an efficient bus for transmitting excitations to the active sensing element. This creative explosion in vdW-based superconducting electronics is rapidly growing, and our review highlights the resulting devices and physics. The confluence of vdW JJs with twistr

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Agentic Microphysics: A Manifesto for Generative AI Safety

arXiv

2026-04-16

Abstract. This paper advances a methodological proposal for safety research in agentic AI. As systems acquire planning, memory, tool use, persistent identity, and sustained interaction, safety can no longer be analysed primarily at the level of the isolated model. Population-level risks arise from structured interaction among agents, through processes of communication, observation, and mutual influence that shape collective behaviour over time. As the object of analysis shifts, a methodological gap emerges. Approaches focused either on single agents or on aggregate outcomes do not identify the interaction-level mechanisms that generate collective risks or the design variables that control them. A framework is required that links local interaction structure to population-level dynamics in a causally explicit way, allowing both explanation and intervention. We introduce two linked concepts. Agentic microphysics defines the level of analysis: local interaction dynamics where one agent's output becomes another's input under specific protocol conditions. Generative safety defines the methodology: growing phenomena and elicit risks from micro-level conditions to identify sufficient mechanisms, det

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RadAgent: A tool-using AI agent for stepwise interpretation of chest computed tomography

arXiv

2026-04-16

Abstract. Vision-language models (VLM) have markedly advanced AI-driven interpretation and reporting of complex medical imaging, such as computed tomography (CT). Yet, existing methods largely relegate clinicians to passive observers of final outputs, offering no interpretable reasoning trace for them to inspect, validate, or refine. To address this, we introduce RadAgent, a tool-using AI agent that generates CT reports through a stepwise and interpretable process. Each resulting report is accompanied by a fully inspectable trace of intermediate decisions and tool interactions, allowing clinicians to examine how the reported findings are derived. In our experiments, we observe that RadAgent improves Chest CT report generation over its 3D VLM counterpart, CT-Chat, across three dimensions. Clinical accuracy improves by 6.0 points (36.4% relative) in macro-F1 and 5.4 points (19.6% relative) in micro-F1. Robustness under adversarial conditions improves by 24.7 points (41.9% relative). Furthermore, RadAgent achieves 37.0% in faithfulness, a new capability entirely absent in its 3D VLM counterpart. By structuring the interpretation of chest CT as an explicit, tool-augmented and iterative reasoning

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MADE: A Living Benchmark for Multi-Label Text Classification with Uncertainty Quantification of Medical Device Adverse Events

arXiv

2026-04-16

Abstract. Machine learning in high-stakes domains such as healthcare requires not only strong predictive performance but also reliable uncertainty quantification (UQ) to support human oversight. Multi-label text classification (MLTC) is a central task in this domain, yet remains challenging due to label imbalances, dependencies, and combinatorial complexity. Existing MLTC benchmarks are increasingly saturated and may be affected by training data contamination, making it difficult to distinguish genuine reasoning capabilities from memorization. We introduce MADE, a living MLTC benchmark derived from {m}edical device {ad}verse {e}vent reports and continuously updated with newly published reports to prevent contamination. MADE features a long-tailed distribution of hierarchical labels and enables reproducible evaluation with strict temporal splits. We establish baselines across more than 20 encoder- and decoder-only models under fine-tuning and few-shot settings (instruction-tuned/reasoning variants, local/API-accessible). We systematically assess entropy-/consistency-based and self-verbalized UQ methods. Results show clear trade-offs: smaller discriminatively fine-tuned decoders achieve the st

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Picometer-Scale Spatial Symmetry Breaking in Active Transmissive Metasurfaces

arXiv

2026-04-16

Abstract. Active transmissive metasurfaces are central building blocks for future compact, cascadable optical systems, enabling the stacking of multiple functional layers for advanced dynamic beam shaping, photonic neural networks, depth sensing, and holography. We present a transmissive electro-optic metasurface based on silicon-on-lithium-niobate, where an array of silicon waveguides with periodic perturbations, individually controlled at the 100 pm scale, supports well-defined high-Q (>2000) guided-mode resonances (GMRs). We incorporate interdigitated push-pull electrodes between subwavelength-spaced GMR elements to locally tune the refractive index in the lithium niobate substrate, thereby shifting the GMR resonance and enabling opposite phase and amplitude modulation between neighboring radiative elements. In a geometrically symmetric metasurface, this effect introduces electro-optic beam splitting via diffraction, with diffraction efficiencies as high as 3%. By introducing controlled passive resonance detuning via 100 pm scale perturbation shifts, we increase the efficiency of amplitude modulation six-fold through geometrical symmetry breaking, achieving amplitude modulation depths of

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The potential of metabolic engineering for sustainable phytosterol production

OpenAlex

2026-04-16

DOI: <https://doi.org/10.1007/s44307-026-00108-9>

Abstract. Phytosterols, essential components pivotal to plant membrane stability and celebrated for their extensive pharmacological benefits, have garnered considerable attention across industries, including food fortification, nutraceuticals, and pharmaceuticals. The escalating demand for phytosterols, fueled by their myriad health advantages, underscores the urgent need for more efficient synthesis methodologies. Among these, metabolic engineering stands out as a promising approach due to its biologically driven process, which operates under stable conditions, thereby enhancing reaction specificity and drastically reducing the production of undesirable by-products. This review consolidates the latest research endeavors focused on enhancing phytosterol accumulation, providing a comprehensive analysis of strategies including gene manipulation, enzyme engineering, metabolic engineering, and the utilization of diverse host organisms such as bacteria, algae, and yeast. We explore recent advancements in phytosterol biosynthesis and engineering, providing a comprehensive overview of the field's current state and examining promising methodologies for future research and applications.

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Abstract. Type 1 diabetes (T1D) is driven by autoimmune destruction of pancreatic β -cells and remains incurable despite major advances in insulin delivery and glucose-monitoring technologies. Transplantation of primary islets or stem cell-derived β -like cells offers a promising route to physiological glycemic control; however, durable engraftment remains limited by complex immune rejection. Unlike classical solid organ transplantation, β -cell replacement in T1D confronts a uniquely intertwined set of immunological barriers, including innate inflammatory activation, adaptive alloimmunity, persistent humoral responses, and recurrent autoimmune memory, further exacerbated by as-yet undefined factors that disrupt the native islet microenvironment. These overlapping effector pathways help explain why single-axis immunosuppressive or physical shielding strategies have not achieved long-term protection. In this review, we synthesize current mechanistic insights into the immune processes that limit β -cell graft survival and organize emerging therapeutic strategies according to the rejection pathways they target. We discuss advances in graft-intrinsic immune engineering, local graft-adjacent immunomodulation, and systemic immune modulation.

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Abstract. The block-and-lock strategy aims to achieve a functional cure for human immunodeficiency virus type 1 (HIV-1) infection by enforcing durable, drug-independent silencing of proviral transcription. Several latency-promoting agents have been described that effectively limit viral reactivation in vitro or in animal models. However, most approaches induce only partial or reversible transcriptional repression and have not yet been translated into safe and effective clinical interventions. This review summarizes the molecular mechanisms underlying block-and-lock strategies and critically evaluates the limitations of current candidate compounds. We highlight recent advances in understanding HIV-1 integration site selection, focusing on the roles of lens epithelium-derived growth factor p75 (LEDGF/p75) and cleavage and polyadenylation specificity factor subunit 6 (CPSF6) in directing proviral integration toward gene-dense, transcriptionally active chromatin. Pharmacological disruption of the LEDGF/p75–integrase interaction by LEDGF/p75 inhibitors (LEDGINs) redirects proviral integration toward less transcriptionally active genomic regions that are more resistant to reactivation. Recent and ongoing clinical trials are evaluating the efficacy and safety of LEDGINs in combination with latency-reversing agents.

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Haploid and Doubled Haploid Platforms for Wheat Improvement: Methods and Applications

OpenAlex

2026-04-13

DOI: <https://doi.org/10.3390/agronomy16080797>

Abstract. Haploid and doubled haploid (DH) technologies are important tools for accelerated wheat (*Triticum aestivum* L.) breeding, enabling the rapid production of fully homozygous lines and increasing the efficiency of genetic analysis of complex traits. This review presents a comprehensive analysis of the main approaches to producing haploid and DH wheat plants, with particular emphasis on androgenesis-mediated and chromosome elimination methods, including wheat-maize hybridisation. The biological basis of androgenesis is discussed in relation to stress-induced microspore reprogramming; however, the primary focus is on the methodological factors determining the efficiency of DH production, including the donor plant genotype, microspore development stage, pretreatment conditions, composition of the induction and regeneration media, and chromosome doubling. However, its widespread application remains limited by pronounced genotypic dependence, low responsiveness of many commercial varieties, albinism, and a lack of universally effective protocols. In contrast, distant hybridisation systems, particularly wheat-maize hybridisation, are generally characterised by greater reproducibility and less

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